

Prove each of the statements in 23-33.

4.6.29. For any odd integer n , $\lceil n^2 / 4 \rceil = (n^2 + 3) / 4$

Proof:

Suppose n is any odd integer.

We must show that the ceiling of n^2 divided by 4

is equal to the sum of $n^2 + 3$ divided by 4.

Definition of odd:

$$n \text{ is odd} \leftrightarrow n = 2k + 1, \text{ for some } k \in \mathbb{Z}$$

Definition of ceiling:

$$\lceil x \rceil = n \leftrightarrow n - 1 \nmid x \leq n$$

Let $k \in \mathbb{Z}$

$$n = 2k + 1$$

$$\lceil n^2 / 4 \rceil = \lceil (2k + 1)^2 / 4 \rceil$$

$$(2k + 1)^2 = 4k^2 + 4k + 1$$

$$(4k^2 + 4k + 1) / 4 = k^2 + k + (1 / 4)$$

$$\lceil n^2 / 4 \rceil = \lceil k^2 + k + (1 / 4) \rceil$$

$$(n^2 + 3) / 4 = ((2k + 1)^2 + 3) / 4$$

$$((4k^2 + 4k + 1) + 3) / 4 = (4k^2 + 4k + 4) / 4 = k^2 + k + 1$$

$$(n^2 + 3) / 4 = k^2 + k + 1$$

$$\lceil k^2 + k + (1 / 4) \rceil = (k^2 + k + 1) \leftrightarrow (k^2 + k + 1) - 1 \nmid k^2 + k + (1 / 4) \leq k^2 + k + 1$$

$$(k^2 + k + 1) - 1 \nmid k^2 + k + (1 / 4) \leq k^2 + k + 1$$

$$k^2 + k \nmid k^2 + k + (1 / 4) \leq k^2 + k + 1$$

The inequality $k^2 + k \nmid k^2 + k + (1 / 4) \leq k^2 + k + 1$ is

true.

Hence, $\lceil k^2 + k + (1 / 4) \rceil = ((2k + 1)^2 + 3) / 4$ is

true, which is what was to be shown.

□

Solution: For any odd integer n , $\lceil n^2 / 4 \rceil = (n^2 + 3) / 4$ is true.